Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Data- and AI-based mobility avatar assistant for sustainable and user-centered mobility*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (X), Secondary Battery ( ), Advanced Mobility (X), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (X), Next-gen Communication (X), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Electrical・Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Digital Content	*Virtual Reality*	*Imaging/Audio Devices*	*Electronic Display Boards*
Concept of Proposed Technology	Core concept: - Development of an AI-based avatar as a mobility assistant to support and optimize individual mobility decisions. - Realization of a context-aware (e.g. day time, city structure, weather, traffic))and user individual (camera based recognition of user types) behavior and recommendations of the assistant, - Integration into public touchpoints (interactive screens at mobility stations, key intersections, points of interest, transportation hubs) offering real-time, location-based mobility information and guidance for travel planning and innovative mobility solutions. - AI-driven integration of mobility-relevant data sources and contextual services along the travel chain (e.g., restaurants, events). - Definition and realization of symbiotic data-exchange strategies with personal avatar- and mobilty-smartphone-apps and evaluation of the official creation of a collaborating user specific smartphone-app with personal all-time-avatar Main functions and potential uses: - Data- and model-based intelligence, using predictive algorithms based on advanced user behavior models by Fraunhofer IAO and real-time data (e.g. traffic, urban infrastructure, weather, POIs) for context-aware mobility and added service recommendations. - Multimodal Route Optimization, supporting seamless travel by integrating multiple transport modes (e.g., walking, cycling, public transport, and private vehicles), optimizing routes according to time, cost, convenience, sustainability aspects and user-defined preferences. - Integration of the Fraunhofer Mobility Hub Planning Tool for improved mobility infrastructure planning - Contextual Service Recommendations: Dynamic, location-based suggestions for restaurants, services, cultural activities, and events along the user's route or at mobility hubs.			

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Rationale for Joint R&D Project	- The needed integration of various data sources (static, dynamic, local information but also user types and behavior patterns) must be based on an interdisciplinary approach - User acceptance must be seen as a critical factor for the adoption of the innovation as the users have to interact with a public infrastructure what needs at first their attention and interest in a short timeframe and afterwards their trust for interacting in the public space. Thus the user acceptance has to be analyzed and defined in a scientific way to guarantee a later market success - The huge variety in the acceptance, use and style of avatars depending on global regions and national behavior patterns, requires an international consortium to ensure a later global market and sales strategy.
Key Words	- AI-based mobility avatar as advising assistant - Multimodal travel guidance - Real-time, context-aware recommendations - Mobility hub planning integration - Interacting public smart screens

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (2/3)	**End of the Project**	Level (7/8)
R&D Objectives	- Behavioral modeling and predictive algorithms for personalized, context-aware travel guidance based on user behavior and preferences - Multimodal routing interface for integration and optimization across multiple transport modes - Real-time data integration framework for mobility, infrastructure, and contextual local data - Interactive public screen interfaces for intuitive, adaptive interaction with stationary avatars - Learning and personalization mechanisms by continuous adaptation to user behavior and preferences.			
R&D Target Specifications	- User Engagement: Target ≥50% interaction rate among passersby at deployed touchpoints during pilot phase - Operational Availability: Avatar system operates with ≥99% uptime during operating hours at mobility touchpoints - Multimodal Coverage: Supports at least 4 transport modes (walking, cycling, public transport, private/shared vehicles) for route planning and increases the acceptance of those through reliable and convenient recommendations that improves the quality and increases the travel time of the users' mobility routines - Recommendation Accuracy: At least 80% of users rate mobility and local service recommendations as relevant and helpful during pilot evaluations - Response Time: Avatar provides answers and recommendations in under 2 seconds per user query - Integration with the Fraunhofer Mobility Hub Planning Tool to support optimized mobility hub planning			
R&D Plan	- Iterative R&D approach combining AI development, data integration, human–avatar interaction, and real-world pilot validation, structured in the following key tasks: - Requirements & system design, incl. user and stakeholder analysis, system architecture and data interface definitions (M1-6) - Core technology development, incl. behavioral modeling and AI algorithms, multimodal routing interface, interactive avatar interface (M7-18) - Integration, pilot deployment and adaptation, incl. integration with real-time mobility and contextual data, deployment at selected mobility touchpoints, field tests, evaluation and system optimization (M19-36) - Development of a commercialization/rollout roadmap (M30-36) - Milestone M6: Finalized system concept and KPIs, Milestone M18: Functional avatar prototype, Milestone M25: Pilot system implemented and ready for testing			

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. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Specification and further development of behavioral models for mobility decision support - Design and evaluation of human–avatar interaction concepts - Algorithmic support for context-aware mobility recommendations - Scientific evaluation, KPI definition, and validation in real-world pilots - Knowledge transfer and documentation
Expected R&R of Korean SME Partner	- Software- and hardware development and Integration - Technical integration and prototyping, including interface definition, system compatibility and joint prototyping of touchpoint solutions - Testing and validation within real-world conditions - Market entry strategy and commercial deployment planning

5. Expected Outcomes/Impact

Expected Outcomes and Impact	- A validated, market-ready stationary mobility avatar for deployment at public mobility touchpoints - Design guidelines (visual and behavior) and archetype concepts for public avatars - Scalable software and interaction modules that can be integrated into existing mobility infrastructure and smart city systems. - Direct applicability for cities, transport authorities, mobility hub operators, and infrastructure providers. - Scalable, modular solution for smart city and mobility infrastructure

6. References

(1) Matowicki, M., Pecherkova, P., Amorim, M., Kern, M., Motzer, N., & Pribyl, O. (2024). Complementing or competing with public transit? Evaluating the parameter sensitivity of potential Mobility-as-a-Service (MaaS) urban users in Germany, the Czech Republic, Poland, and the United Kingdom with a mixed choice model. *Transportation*, 1-34.

(2) Tsaga, I., Motzer, N., Kern, M., & Amorim, M. (2023). A mobility app for everyone? An empirical analysis of user types based on social lifestyles and personal characteristics. *Human Factors in Transportation*, *95*(95).

(3)Kern, M., Kluge, N., Burkard, B., Amorim, M., & Motzer, N. (2023). Acceptance analysis of value-creating service cooperations and extensions for Mobility-on-Demand solutions. *Human Factors in Transportation*, *95*(95)

(4) Matowicki, M., Amorim, M., Kern, M., Pecherkova, P., Motzer, N., & Pribyl, O. (2022). Understanding the potential of MaaS–An European survey on attitudes. *Travel Behaviour and Society*, *27*, 204-215.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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